

Sheet 4

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To be discussed on Tuesday, October 13, 16:00

Submission deadline: Monday, October 12, 23:59 (Basel time)

Please write your full name and student ID number directly on the assignment. **We accept only handwritten solutions.**

Conventions. All vector spaces are finite-dimensional over $\mathbb{C}$. For two qubits, use the ordered basis $(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$, where $|xy\rangle = |x\rangle \otimes |y\rangle$ and the first label refers to the first qubit. The single-qubit matrices used below are

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

1. Spectral Projectors and Pauli Rotations

Let O be a Hermitian operator whose only eigenvalues are $+1$ and -1 .

- (a) Derive the spectral projectors $P_{\pm} = (I \pm O)/2$. Verify that they are orthogonal, i.e.,

$$P_{\pm}^2 = P_{\pm}, \quad P_{\pm}^{\dagger} = P_{\pm}.$$

- (b) Write six spectral projectors in ket-bra form of X , Y and Z .

- (c) Let the pure qubit state $|\psi\rangle$ correspond to the Bloch vector $\vec{r}$. Determine the Bloch vectors corresponding to $X|\psi\rangle$, $Y|\psi\rangle$, and $Z|\psi\rangle$. Give a geometric interpretation of the action of each Pauli matrix on the Bloch sphere.
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2. Spectral Decomposition and a Function of an Operator

In the computational basis $(|0\rangle, |1\rangle)$, consider

$$A = \begin{pmatrix} 2 & 1-i \\ 1+i & 3 \end{pmatrix}.$$

- (a) Verify that A is Hermitian. Find its eigenvalues $\lambda_1 < \lambda_2$ and corresponding normalised eigenvectors $|v_1\rangle, |v_2\rangle$. Check that these vectors form an orthonormal basis.

To construct the change-of-basis matrix U , place these eigenvectors, expressed in the computational basis, in its columns:

$$U = (|v_1\rangle \ |v_2\rangle).$$

Construct $D = \text{diag}(\lambda_1, \lambda_2)$, keeping the eigenvalues in the same order as the columns of U . Verify that $U^{\dagger}U = I_2$ and $U^{\dagger}AU = D$.

Meaning of the change of basis. If x and c are the coordinate columns of the same vector in the computational basis and the eigenbasis, respectively, then

$$x = Uc, \quad c = U^{\dagger}x,$$

where U is unitary. Thus D represents the same operator as A , but in its eigenbasis, where each coordinate is multiplied by the corresponding eigenvalue.

- (b) Construct the spectral projectors P_λ as outer products and as matrices. Write the spectral decomposition $A = \sum_\lambda \lambda P_\lambda$. Find $\text{tr } A$ from the eigenvalues and explain why this works.
- (c) Is A positive definite? Use its spectral decomposition to find the positive square root $B = \sqrt{A}$, both in projector form and as a matrix.

3. What the Spectral Theorem Does and Does Not Say

Terminology. A matrix M is *diagonalisable by similarity* if $S^{-1}MS$ is diagonal for some invertible S . It is *unitarily diagonalisable* if $U^\dagger MU$ is diagonal for some unitary U . The columns of S form an eigenbasis; for a unitary U , that eigenbasis must also be orthonormal.

Consider

$$M = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad N = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}.$$

- (a) Compute $M^\dagger M$ and $MM^\dagger$. Is M normal? Find the eigenvalues and eigenspaces of M . Is it diagonalisable by similarity?
- (b) Find the eigenvalues and eigenvectors of N , and an invertible S for which $S^{-1}NS$ is diagonal. Are its two eigenspaces orthogonal? Check whether N is normal by comparing $N^\dagger N$ and $NN^\dagger$.
- (c) State the spectral theorem for complex matrices and use it to decide whether either matrix is unitarily diagonalisable. Explain why the result for N in part (b) does not contradict the theorem. Also check $N^2 = I_2$: does this identity make N unitary?

4. Tensor Products, CNOT and Entanglement

Background: product states and entanglement.

All states considered in this exercise are *pure states*, described by normalised state vectors. A general pure state of two qubits is

$$|\psi\rangle = c_{00}|00\rangle + c_{01}|01\rangle + c_{10}|10\rangle + c_{11}|11\rangle, \quad \sum_{x,y=0}^1 |c_{xy}|^2 = 1,$$

where $|xy\rangle = |x\rangle \otimes |y\rangle$.

The state is called a *product state* if it can be written as

$$|\psi\rangle = |a\rangle \otimes |b\rangle,$$

where $|a\rangle$ and $|b\rangle$ are normalised single-qubit states. Thus the joint state can be specified by assigning a pure state to each qubit separately.

Writing

$$|a\rangle = a_0|0\rangle + a_1|1\rangle, \quad |b\rangle = b_0|0\rangle + b_1|1\rangle,$$

and expanding their tensor product gives

$$|a\rangle \otimes |b\rangle = a_0 b_0 |00\rangle + a_0 b_1 |01\rangle + a_1 b_0 |10\rangle + a_1 b_1 |11\rangle.$$

Therefore, to test whether a state is a product state, try to find single-qubit coefficients satisfying

$$c_{00} = a_0 b_0, \quad c_{01} = a_0 b_1, \quad c_{10} = a_1 b_0, \quad c_{11} = a_1 b_1.$$

If no such factorisation exists, the pure state is called *entangled*. To establish entanglement, show that the coefficient equations cannot all be satisfied.

A superposition of several computational-basis vectors can still be a product state. For example,

$$\frac{|00\rangle + |01\rangle}{\sqrt{2}} = |0\rangle \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Here the first qubit is in $|0\rangle$ and the second is in $(|0\rangle + |1\rangle)/\sqrt{2}$, so the state is not entangled. The relevant question is whether the joint state factorises, rather than how many terms appear in its basis expansion.

- (a) Compute the 4×4 matrices $X \otimes I_2$ and $I_2 \otimes X$ and describe which qubit each flips. Show, using the mixed-product rule, that these different operators commute.
- (b) The controlled-NOT gate, with the first qubit as control and the second as target, is

$$C = \text{CNOT} = |0\rangle\langle 0| \otimes I_2 + |1\rangle\langle 1| \otimes X.$$

Determine C 's action on all four computational basis vectors and write its matrix.

- (c) Decide which of the following normalised pure states are product states and which are entangled. Factor any product state:

$$|\chi\rangle = \frac{|00\rangle - |01\rangle + |10\rangle - |11\rangle}{2}, \quad |\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

- (d) Compute $C(H \otimes I_2)|00\rangle$. Show that an operator of the form $A \otimes B$ maps a product vector to another product vector, whenever the output is nonzero. Use this to explain why C cannot be written as $A \otimes B$.

5. Hadamards on an n -Qubit Register

Let $n \geq 1$ and label the computational basis by $i, j \in \{0, \dots, 2^n - 1\}$. Use the binary convention

$$i = \sum_{k=1}^n i_k 2^{n-k}, \quad |i\rangle = |i_1\rangle \otimes \dots \otimes |i_n\rangle, \quad i_k \in \{0, 1\},$$

and similarly for j .

- (a) Show that

$$\langle i | H^{\otimes n} | j \rangle = \frac{(-1)^{i \cdot j}}{\sqrt{2^n}}, \quad i \cdot j = \sum_{k=1}^n i_k j_k \pmod{2}.$$

Hint: First express the four single-qubit matrix entries $\langle a | H | b \rangle$ in one formula, with $a, b \in \{0, 1\}$.

- (b) Using the expressions from (a), find $H^{\otimes n} |0\rangle^{\otimes n}$. How many computational-basis amplitudes are nonzero, and what are the corresponding measurement probabilities? Does this operation create entanglement from $|0\rangle^{\otimes n}$? Justify your answer by writing the output as a tensor product.